

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			frames shape memory alloy / SMA - regains shape after bending both needed for (1) lenses photochromic pigment) - changes colour with changing light (intensity) / sunlight both needed for (1) do not accept sun award (1) for both names if both properties incorrect	2			2		
	(b)	(i)		transparent rather than white /opaque (when applied) accept clear rather than white	1			1		
		(ii)		can / could pass through the skin / get into bloodstream / get into the body (1) <u>long-term</u> effect is <u>unknown</u> / <u>could be</u> toxic build-up <u>over time</u> (1) neutral answer - toxic / poisonous	2			2		
		(iii)		10^3 / 1000 (2) accept 1.2×10^3 / 1200 if answer is incorrect award (1) for $\frac{3 \times 10^{-7}}{2.5 \times 10^{-10}}$		2		2	2	
				Question 5 total	5	2	0	7	2	0